Data Dashboard Mirage Classroom Run Kit

TOK Cognitive Science Activity Bank • Knowledge and Technology • 30-40 min

Category	Knowledge and Technology	**Time**	30-40 min
Difficulty	Medium	**ID**	data-dashboard-mirage

Big idea: Dashboards do not simply show data; they prioritize, compress, and narrate it.

One-Minute Teacher Brief

Students compare two dashboards built from the same data and evaluate how interface design shapes conclusions.

Student Prompt

Inspect Data Dashboard Mirage Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Data Dashboard Mirage Stimulus A	Student-facing stimulus 1: Dashboard A highlights falling late arrivals in green; Dashboard B highlights rising absences in amber using the same school-week dataset. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Data Dashboard Mirage Stimulus B	Student-facing stimulus 2: Dashboard A ranks classes by average score; Dashboard B ranks them by improvement. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Data Dashboard Mirage Reveal / Twist	Student-facing stimulus 3: Dashboard A ranks classes by average score; Dashboard B ranks them by improvement. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — Dashboards do not simply show data; they prioritize, compress, and narrate it.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Data Dashboard Mirage Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Data Dashboard Mirage Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	When does data visualization become argument?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Technology.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how dashboards do not simply show data; they prioritize, compress, and narrate it.

• Strong response: separates observation from inference.

• Strong response: names a method, source, model, language choice, context, or perspective.

• Strong response: revises confidence when new evidence or a new criterion appears.

• Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

• Work silently for the first response so you can notice your own starting point.

• Record one knowledge claim, the evidence behind it, and your confidence level.

• After the reveal or comparison, update your claim in a different colour or column.

• Use the debrief to answer: When does data visualization become argument?

Setup Checklist

• Use the same fictional dataset in both dashboard versions.

• Change only visual design choices such as color, order, scale, and highlighted metric.

Example Stimuli or Materials

• Dashboard A highlights falling late arrivals in green; Dashboard B highlights rising absences in amber using the same school-week dataset.

• Dashboard A ranks classes by average score; Dashboard B ranks them by improvement.

Resources To Use

• Resource pack: resources/data-dashboard-mirage/index.html

• Card deck CSV: resources/data-dashboard-mirage/cards.csv

• Visual prompt: resources/data-dashboard-mirage/visual-prompt.svg

• Dashboard audit checklist: title, color, scale, missing context, default ordering, highlighted number.

• Redesign template for a more transparent dashboard.

Run Sequence

1. Show students two dashboards with the same underlying data but different layouts.
2. Groups identify the story each dashboard seems to tell.
3. Students trace which design choices create that impression.
4. Groups redesign one dashboard for greater transparency.

5. Debrief how interfaces shape what users think they know.

Debrief Moves

• Knowledge question: When does data visualization become argument?

• Can a dashboard be accurate and still misleading?

• What should designers owe users who rely on visualized data?

Teacher Quick Script

• A dashboard answers before we know what question it was built to answer.

• Which design choice did the most interpretive work?

Exit Ticket

• One thing I first treated as evidence was...

• My confidence changed because...

• A better knowledge question I can now ask is...